

DATA-DRIVEN INSIGHTS: SEGMENTING AND PREDICTING TOURIST SPENDING BEHAVIOR FOR OPTIMIZED MARKETING

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Abstract

Tourism is a cornerstone of the global economy, significantly contributing to job creation, foreign exchange earnings, and cultural exchange. However, traditional approaches to segmenting and predicting tourist spending behavior often fail to capture the complexity of modern tourism dynamics. This study leverages data mining techniques to enhance tourism marketing by integrating segmentation and predictive modeling. Using a sample size of approximately 20,000 tourists around the globe from UAE and Tanzania, the research identifies spending patterns, preferences, and key determinants of expenditure behavior. The findings aim to provide actionable insights for optimizing resource allocation, improving customer satisfaction, and developing targeted marketing strategies. By bridging the gap between advanced data analytics and tourism marketing, this study underscores the potential of data-driven approaches to foster more effective and sustainable tourism initiatives.

Keywords: *tourist spending behavior, data mining, segmentation and prediction, machine learning in tourism, cost-sensitive learning*

1. INTRODUCTION

Tourism is widely regarded as one of the most dynamic and influential sectors of the global economy, contributing significantly to economic growth, job creation, and the development of various regions. As the tourism industry continues to evolve, the need for effective strategies to cater to the diverse preferences of travelers has become increasingly important. Understanding the factors that drive tourist spending behavior is crucial for businesses, governments, and tourism boards to provide better services, enhance customer satisfaction, and maximize the sector's economic impact. Traditional methods of segmenting tourists have primarily relied on demographic and psychographic data, focusing on basic characteristics such as age, income, nationality, and travel motivations. While these methods have proven useful in certain contexts, they often fail to capture the complexity of modern tourist behavior, which is influenced by a multitude of factors, including personal preferences, past experiences, and external influences such as social media (Vanlalhriati et al., 2019) or global events (Zlatanov, 2015). Additionally, such traditional segmentation techniques tend to rely on relatively small or static datasets, which may not fully reflect the rapidly changing and diverse nature of the global tourist population.

In previous work, Sriphaew & Sombatsricharoen (2015) proposed an innovative approach to recommendation systems using a modified ant colony algorithm, which was successfully applied to the tourism sector. This method highlighted the importance of utilizing intelligent algorithms to better understand user behavior and enhance the personalized travel experience. Moreover, Sriphaew & Katkaeo (2017) explored online hotel reservation platforms and their usability, shedding light on how online behavior influences travel decisions. This study aligns with the need for data-driven insights in understanding customer preferences, an area that this research seeks to further investigate within the tourism context.

In recent years, advancements in data collection and analysis, particularly with the advent of big data and machine learning, have enabled researchers and practitioners to gain deeper insights into tourist behavior, Sung et al. (2021). Large-scale datasets, including transaction records, geotagged data, and online reviews, provide rich sources of information that can reveal patterns and trends previously inaccessible through conventional methods. These datasets, when analyzed using sophisticated techniques such as data mining, allow for the identification of complex relationships between various factors that influence tourist spending, including demographic characteristics, travel preferences, and spending behaviors.

This study seeks to bridge the gap between traditional segmentation methods and the emerging field of data-driven marketing by employing data mining techniques to segment tourists based on their spending behavior. Specifically, the study focuses on identifying and predicting tourist spending patterns, offering a more nuanced and data-driven approach to tourism marketing. The core research question of this paper is: How can data mining techniques be applied to segment tourists and predict their spending behavior, thereby enhancing the ability to target marketing efforts effectively? Building on prior work in the application of data-driven methods, such as in Sriphaew et al. (2022) on consumer decision factors and the integration of machine learning techniques in tourism contexts, this paper further explores actionable insights through predictive modeling. By integrating segmentation and predictive modeling, this study aims to uncover insights that can assist in the development of more tailored and effective marketing strategies. The ultimate goal is to optimize the allocation of marketing resources, increase customer satisfaction, and ensure that tourism businesses can meet the diverse needs of their customers more effectively.

2. LITERATURE REVIEW

Existing Approaches to Tourist Segmentation and Prediction Tourist segmentation has evolved significantly over the years. Early efforts relied on basic demographic and psychographic profiles, but recent studies have shifted toward more data-driven techniques that leverage big data and machine learning. These methods allow for more precise and dynamic segmentation, offering a deeper understanding of consumer behavior.

Tourism Profile Measure for Data-Driven Segmentation Guérard et al. (2024) explored the use of geotagged data and social media activity to segment tourists using clustering algorithms. Their research demonstrated how data-driven segmentation could be applied to urban destinations, providing valuable insights into travel behavior. By leveraging digital traces, their study enhanced the granularity of segmentation, enabling targeted marketing strategies for diverse tourist groups.

In line with the advancements in data-driven approaches, Sriphaew et al. (2021) explored the application of machine learning techniques in predicting consumer decisions and behaviors across various domains, including tourism. Their study on digital and physical video game purchases (Napakornwattananon & Sriphaew, 2022) demonstrated the power of predictive models in understanding complex consumer choices, a principle that can be directly applied to understanding tourist spending behaviors in this research.

Machine Learning in Sustainable Tourism Núñez et al. (2024) employed machine learning techniques, such as Decision Trees and Random Forests, to predict tourist spending patterns in Saudi Arabia. Their study integrated these models with ARIMA forecasting, aiming to align predictions with sustainable tourism objectives. The integration of machine learning into sustainable tourism highlights the potential of predictive models in optimizing resource allocation while maintaining sustainability.

Building on the work of Sriphaew & Sombatsricharoen (2015) and Sriphaew & Katkao (2017), who both utilized computational models to enhance decision-making in tourism and hospitality contexts, this study explores the intersection of predictive analytics with sustainable tourism practices. Specifically, it focuses on how machine learning models can be used to balance profitability with environmental and societal concerns in tourism marketing.

While existing studies have made significant strides in the use of clustering and prediction models in tourism, they often fall short in combining both approaches for actionable insights. Furthermore, many studies rely on localized datasets, such as social media activity or geotagged information, which may not reflect the global diversity of tourist behavior. Additionally, few studies have integrated cost-sensitive learning into predictive modeling, which could provide more practical business applications by prioritizing high-value customers.

This research aims to address these gaps by:

- Integrating segmentation and predictive modeling to derive actionable insights from tourist spending patterns.
- Utilizing structured datasets that incorporate demographic, behavioral, and spending attributes to enhance the accuracy of predictions.
- Applying cost-sensitive learning to align model predictions with marketing objectives, thus providing more relevant insights for tourism businesses.

3. METHODOLOGY

3.1 Dataset

This study utilizes two distinct datasets to analyze and classify tourist behaviors and expenditures. The combination of these datasets aims to reduce country-specific bias and increase the overall data size, providing a more comprehensive foundation for analysis.

- 1) *Tourism Behavior Analysis in UAE Dataset*: This dataset contains detailed information on the behaviors and preferences of tourists visiting the United Arab Emirates (UAE). It includes 1,000 records and 15 features such as tourist demographics, travel purpose, spending patterns, accommodation types, and travel frequency (Aliriaz, 2023).
- 2) *Tanzania Tourism Classification Challenge Train Dataset*: This dataset focuses on classifying tourists' expenditure ranges in Tanzania. It includes 18,506 records and 21 features, categorizing expenditures into different cost brackets (Temu, 2023). The dataset highlights the economic significance of the tourism sector in Tanzania and includes various classification labels for tourist spending.

To overcome the limitations of country-specific data and enhance the robustness of the analysis, two datasets were combined: the Tourism Behavior Analysis in UAE Dataset (Aliriaz, 2023) and the Tanzania Tourism Classification Challenge Train Dataset (Temu, 2023). This integration resulted in a larger and more diverse dataset, allowing for a broader analysis of tourist behavior patterns and expenditure classifications across different regions, (Puttanapong et al., 2024).

With a dataset of integrated 19,506 records, the final integrated dataset comprises 11 attributes that capture key demographic and travel-related information. These attributes are: *touristID*, *age_group*, *gender*, *country of origin*, *travel_purpose*, *preferred_destination*, *travel_frequency_per_year*, *travel_with*, *total_spending_USD*, *stay_duration_days*, and *spending_type*.

The target variable, *Spending Type*, was derived by categorizing tourists based on their total expenditure. For the Tourism Behavior Analysis in UAE Dataset, the target variable was derived from *Total_Spending_USD* by a discretization technique with equal-width binning. It divided the range of total expenditures into intervals of equal size (Han et al., 2012). Based on this method, travelers were segmented into three categories:

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|------------------------------------|--|
| - <i>High-Spending Travelers</i> | (>3400 <i>Total_Spending_USD</i>) |
| - <i>Medium-Spending Travelers</i> | (1800-3400 <i>Total_Spending_USD</i>) |
| - <i>Low-Spending Travelers</i> | (<1800 <i>Total_Spending_USD</i>) |

This categorization enables a more nuanced understanding of spending behavior and supports the development of tailored strategies for various traveler types.

For the Tanzania Tourism Classification Challenge Train Dataset, the target variable was already defined in the *cost_category* feature. However, there are 6 distinct values in this feature, i.e., high cost,

higher cost, highest cost, normal cost, low cost, lower cost. Therefore, the target variable is redefined by combining into three categories:

- *High-Spending Travelers* (high cost, higher cost, and highest cost)
- *Medium-Spending Travelers* (normal cost)
- *Low-Spending Travelers* (low cost, lower cost)

There is no concern about class imbalance, as the distribution is well-balanced across the three categories with 9,219 high, 5,809 medium, and 4,478 low.

3.2 Data Preprocessing and Model Development

The CRISP-DM (Cross-Industry Standard Process for Data Mining) framework was employed to guide the development of the predictive model for analyzing tourism spending patterns (Shearer, 2000). This structured methodology helped in ensuring a systematic approach to data preprocessing, model training, and evaluation. The CRISP-DM process is broken down into several critical phases, each addressing key components of the workflow.

1) Data Cleaning

- *Merging and Reconciling Datasets:*

The integration of two tourism datasets from distinct geographical regions resulted in a larger and more diverse dataset. Discrepancies between column names and data formats were resolved to ensure consistency, facilitating a seamless combination of the two datasets. This step was essential in minimizing biases associated with country-specific data and allowed for a more generalized analysis of tourist behavior.

- *Handling Missing Values:*

To maintain data integrity, missing values in continuous variables were addressed through imputation. Specifically, missing values in the `Travel_Frequency_per_Year` column were replaced with the median value, reflecting the central tendency of the dataset. For categorical variables, missing entries were imputed with the placeholder "Unknown" to preserve the completeness of the dataset.

- *Outlier Management:*

Outliers in the `total_cost_USD` column were filtered to ensure that extreme values did not disproportionately influence the analysis. The spending data was restricted to a range between 100 and 15,000 USD to remove outlier values that could skew the model's performance.

2) Preprocessing

- *Encoding:*

Categorical features such as `Travel_Purpose` and `Accommodation_Type` were encoded using `LabelEncoder`. This transformation allowed these categorical variables to be converted into numerical format, which is required for machine learning models.

- *Scaling:*

Numerical features, including `Age` and `Travel_Frequency_per_Year`, were standardized to have a mean of zero and a standard deviation of one. Scaling these features ensured that they contributed equally to the model, preventing any feature with a larger range from dominating the learning process.

- *Balancing:*

Given the class imbalance in the target variable `Traveler_Type`, which classifies travelers into High, Medium, and Low spending categories, oversampling techniques were applied to address the skew in class distribution. This helped ensure that each category was adequately represented, reducing bias in the classification model.

3) Model Training and Evaluation

- **Baseline Model:**

A Random Forest classifier was initially used as a baseline model. The model achieved an accuracy of 74.65% after preprocessing. This result provided a solid starting point but indicated that further model optimization was necessary to improve classification performance.

- **Tuned Model:**

The Random Forest model was further optimized by increasing the number of trees to 300 and setting the maximum depth of each tree to 18. This adjustment resulted in a modest improvement in performance, achieving a cross-validation accuracy of 75.57%. The tuned model demonstrated enhanced capability in classifying tourist spending categories.

- **Cost-Sensitive Learning:**

To align model predictions with business objectives, a cost matrix was incorporated into the training process. The matrix adjusted the model's classification approach, penalizing misclassifications of high-value travelers more heavily, which is important for ensuring that marketing and business strategies are appropriately prioritized.

By following the CRISP-DM framework, a structured and systematic approach was applied to each phase of the project. From merging and cleaning the data to preprocessing and final model evaluation, this methodology ensured that the data mining process was rigorous, leading to a robust and effective model for predicting tourism spending behavior.

4. RESULT AND DISCUSSION

Table 1 summarizes the empirical results of the Random Forest classification model applied to predict tourist spending behavior, comparing the performance of the Baseline Model, Tuned Model, and the Cost-Sensitive Learning Model.

Table 1. **Empirical Results of Random Forest Classification**

Method	Accuracy	Class	Precision	Recall	F1-Score	F1-Score Average
Baseline Model	74.7%	High	76%	75%	76%	75%
		Medium	70%	70%	70%	
		Low	78%	79%	79%	
Tuned Model	75.6%	High	80%	76%	78%	75.3%
		Medium	66%	75%	70%	
		Low	81%	76%	78%	
Cost-Sensitive Learning	75.0%	High	80%	72%	76%	75.3%
		Medium	65%	78%	71%	
		Low	83%	76%	79%	

Baseline Model Performance

The Baseline Model (standard Random Forest classifier) achieved an overall accuracy of 74.7%. It showed relatively balanced precision and recall across the three spending categories, with its best performance in the low-spending category (F1-Score of 79%). However, the model struggled with the medium-spending category, where both precision and recall were 70%, leading to a lower F1-Score of 70%. This suggests that while the model was effective in predicting low-spending tourists, it was less reliable for medium-spending travelers.

Tuned Model Performance

By tuning the model parameters—specifically, increasing the number of trees to 300 and setting the maximum depth of each tree to 18—the Tuned Model showed a slight improvement in overall accuracy

(75.6%). The most significant improvement was in the high-spending category, where precision and recall increased to 80% and 76%, respectively, leading to an F1-Score of 78%. The medium-spending category saw less improvement, with precision at 66% and recall at 75%, resulting in an F1-Score of 70%. However, the low-spending category remained strong, maintaining an F1-Score of 78%.

Cost-Sensitive Learning Model Performance

Introducing Cost-Sensitive Learning to prioritize high-value tourists (high-spending) yielded promising results. The overall accuracy remained close to 75.0%, but precision for the high-spending category improved to 80%, while recall decreased slightly to 72%, resulting in an F1-Score of 76%. In the medium-spending category, precision dropped to 65%, but recall increased to 78%, resulting in an F1-Score of 71%. The low-spending category showed the best performance, with precision reaching 83% and F1-Score at 79%, indicating that the model was particularly effective in predicting low-spending tourists.

Segmentation and Marketing Implications

The segmentation into high, medium, and low spending categories provides actionable insights for crafting tailored marketing strategies that align with different customer needs. Drawing from Lappalainen et al. (2021), who emphasized the importance of customizing business approaches based on segmentation in the context of B2B sales, similar principles can be applied to tourism marketing, where understanding distinct customer segments enables more effective marketing efforts.

- *High-Spending Tourists:* High-Spending Tourists: With improved precision and recall, this segment can be targeted with luxury accommodations, exclusive experiences, and personalized services. Marketing efforts should focus on premium brand partnerships, customized packages, and loyalty programs that emphasize exclusivity.
- *Medium-Spending Tourists:* This group requires offerings that balance comfort with affordability. Social media marketing should highlight visually appealing, cost-effective experiences, seasonal promotions, and bundled packages that offer value while encouraging longer stays or family bookings.
- *Low-Spending Tourists:* The focus for this group should be on budget-friendly accommodations, cost-effective tours, and family-oriented deals. Collaborations with local businesses to offer discounts and promotions will further enhance the affordability aspect of the travel experience.

The results reveal that the Tuned Model provided a more balanced performance across all spending categories, with particular improvements in the high and low-spending segments. However, the Cost-Sensitive Learning Model showed superior performance in targeting high-value customers, as penalizing misclassifications allowed for prioritizing high-spending tourists without sacrificing accuracy in medium and low-spending segments. This strategy, inspired by B2B segmentation and targeting strategies (Lappalainen et al., 2021), enables tourism businesses to focus on high-value customers while maintaining reasonable accuracy across all groups. By adopting such a model, businesses can enhance their marketing efficiency and better allocate resources to meet the needs of their diverse customer base.

5. CONCLUSION

This study demonstrates the potential of data-driven approaches to optimize tourism marketing by segmenting tourists based on their spending behavior using machine learning. By applying a Random Forest classifier, we were able to categorize tourists into high, medium, and low spending groups with a cross-validation accuracy of 75.57%. The model's effectiveness was enhanced by using cost-sensitive learning to prioritize high-value tourists, providing actionable insights for marketing strategies. These findings highlight the importance of integrating data mining techniques for more precise, dynamic segmentation, which can lead to better resource allocation, increased customer satisfaction, and improved marketing outcomes. Future research could further enhance predictive capabilities through real-time data and advanced machine learning methods.

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